

Business Understanding Report: Employee Attrition Analysis

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1. Introduction to Employee Attrition

Employee attrition refers to the gradual reduction of a workforce as employees leave an organization and are not immediately replaced. It is typically categorized into voluntary attrition (when an employee chooses to resign) and involuntary attrition (when the company terminates employment). In the context of organizational health, a high rate of voluntary attrition is a critical red flag. It often indicates underlying issues with company culture, compensation, management, or work-life balance. Understanding the root causes of why employees choose to leave is the first step toward building a stable, engaged, and productive workforce.

2. Business Impacts of Employee Attrition

High employee turnover has severe financial and operational consequences for a business.

Direct Financial Costs: Replacing an employee is expensive. Companies must spend money on job advertisements, recruiter fees, interviewing processes, and on boarding materials. It is estimated that replacing an employee can cost anywhere from one-half to two times their annual salary.

Loss of Productivity: When an employee leaves, their workload is often shifted to remaining team members, leading to burnout. Furthermore, a new hire takes months to reach the productivity level of their predecessor.

Loss of Institutional Knowledge: Long-term employees hold valuable knowledge about company processes, client relationships, and historical data. When they leave, this knowledge walks out the door with them.

Impact on Morale: Frequent departures can create a sense of instability, lowering the morale and engagement of the remaining staff and potentially triggering a domino effect of further resignations.

3. Role of Data Science in Strategic HR Decision-Making

Historically, Human Resources has been a reactive, administrative function. Data science transforms HR into a strategic, proactive partner. By leveraging HR analytics, organizations can:

Identify Patterns: Exploratory Data Analysis (EDA) helps uncover hidden correlations, such as whether specific departments, age groups, or salary bands are more prone to leaving.

Shift from Reactive to Proactive: Instead of conducting exit interviews after an employee has already decided to leave, data science allows HR to identify "flight risks" while they are still employed.

Inform Policy Changes: Data-backed insights allow management to make targeted interventions. For example, if data shows that overtime is the primary driver of attrition in the Sales department, HR can implement targeted work-life balance policies rather than applying a blanket, ineffective company-wide raise.

Predictive Modelling: Ultimately, historical data can be used to train machine learning models that predict the probability of a specific employee leaving, allowing HR to intervene early.

4. Defined Business Problem: ABC Manufacturing Ltd

ABC Manufacturing Ltd is currently experiencing employee turnover that is causing concern within the HR department. Before the company invests time and resources into building complex predictive machine learning models, they must first understand the fundamental nature of their workforce and the basic drivers of attrition.

The core business problem is to investigate why employees are leaving the company. As a Junior Data Scientist at AnalystLab Africa Consulting, my objective is to perform an initial exploratory analysis on the company's historical HR data. By answering key business questions—such as the impact of age, income, overtime, and department on turnover—this report will provide the foundational insights required to guide future retention strategies and set the stage for advanced predictive modelling.